

Ekaansh Patel

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EDUCATION

University of Michigan, College of Literature, Science, and The Arts

Expected May 2027

B.S. in Computer Science, GPA: 3.6

Ann Arbor, MI

Awards: Fall 2024 and Fall 2025 University Honors List

Coursework: Data Structures and Algorithms, Computer Science Pragmatics, Introduction to Computer Organization, Foundations of Computer Science

TECHNICAL SKILLS

Languages: C++, Python, Java, HTML, CSS, JavaScript

Libraries/Frameworks: React, Tailwind CSS, Playwright

Developer Tools: VS Code, Git, Linux/Unix, npm, Vite

PROJECTS

Formula 1 Race Analytics Platform

- Engineered a Formula 1 dashboard using React, Vite, and Tailwind CSS, featuring a custom timeline scrubber to seamlessly replay driver track positions, dynamic vehicle metrics (speed, RPM, gear), and live-updating interval timing screens.
- Developed a robust data processing pipeline in Python, utilizing Pandas and NumPy to parse and aggregate massive race datasets via the FastF1 and OpenF1 APIs.
- Designed an interactive strategy module that maps tire compound history and pit stop windows onto dynamic line graphs, visualizing complex team strategies under various track conditions.
- Implemented responsive UI components that synchronize with the playback timeline to display historical weather data and trigger contextual on-screen alerts for critical race events, including safety cars and red or yellow flags.

Drone Delivery

- Engineered a Minimum Spanning Tree utilizing Prim's algorithm to optimize network connectivity across all drone delivery locations.
- Developed a fast heuristic Traveling Salesperson Problem (TSP) solver via the Farthest Insertion algorithm to rapidly approximate efficient flight paths.
- Implemented an optimal TSP solver using a Branch and Bound algorithm, guaranteeing the absolute shortest delivery route by aggressively pruning suboptimal paths.
- Built an interactive web dashboard using HTML, JavaScript, and CSS to visualize complex map data, specific delivery locations, and exact drone flight paths.

Machine Learning - Classifier

- Built a text classification system to automatically identify the subject of course discussion board posts based on post content.
- Applied NLP and machine learning techniques using efficient data structures (sets and maps) to extract and manage word and label statistics.
- Trained the model on historical Piazza posts and implemented a log-likelihood-based classifier to predict the most probable label for each post.

Text Editor

- Created a basic text editor where you are able to type, delete, and move around between text and different lines of text
- Uses abstract data types, dynamic memory, iterators, and linked lists for efficiency in order to move around the different text data

PROJECT TEAMS & ACTIVITIES

Michigan Mars Rover Team (Mrover) | Teleoperations Software Subteam

September 2025 – Present

- Engineered responsive UI features for the online base station, implementing Playwright end-to-end test suites to validate functionality and prevent regressions.
- Enhanced the operator experience by redesigning the data interface, transitioning from legacy data tables to intuitive, clickable components mapped directly onto a visual model of the rover.

Wolverine Sports Analytics

September 2025 – Present

- Collected and scraped player statistics from multiple sports websites, normalized the data into CSV datasets, and implemented a fantasy football projection engine that estimates waiver-wire player point totals using a mathematical model incorporating player usage and historical performance.